

Fig S1

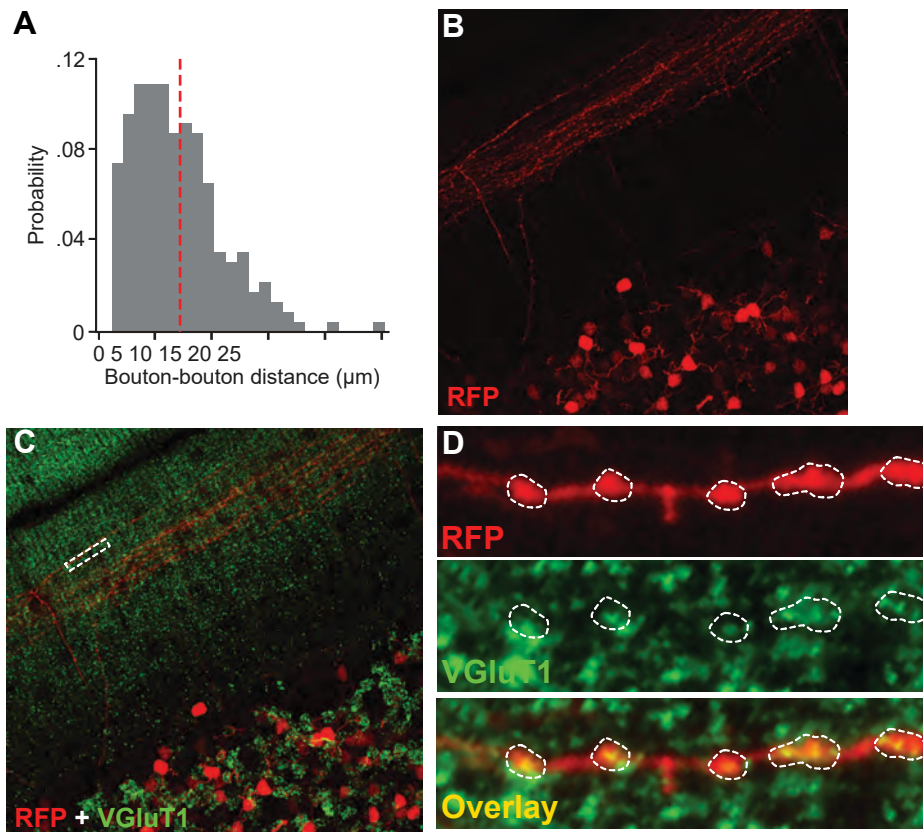


Fig. S1: Most PF varicosities are presynaptic boutons.

A. Histogram of inter-bouton distances measured between consecutive PF enlargements. Mean $\pm$ SD: 7 ± 3.9 μm , red dashed line; $n = 230$ boutons, $N = 22$ mice. **B.** Single-cell-resolved expression of a red fluorescent protein, RFP, in a sparse GC subset using AAV1-syn-Chronos-tdTomato. **C.** Same field of view as in B, with overlay of VGlut1 immunoreactivity, green. The dashed white rectangle highlights the magnified region shown in D. **D.** Magnified PF segment from C, showing RFP-labeled enlargements, top, dashed outlines; VGlut1-positive puncta, middle; and their overlay, bottom. Most PF enlargements co-localize with VGlut1, identifying them as putative synaptic boutons: 93.5%, $n = 43$ enlargements, $N = 2$ mice.

Fig S2

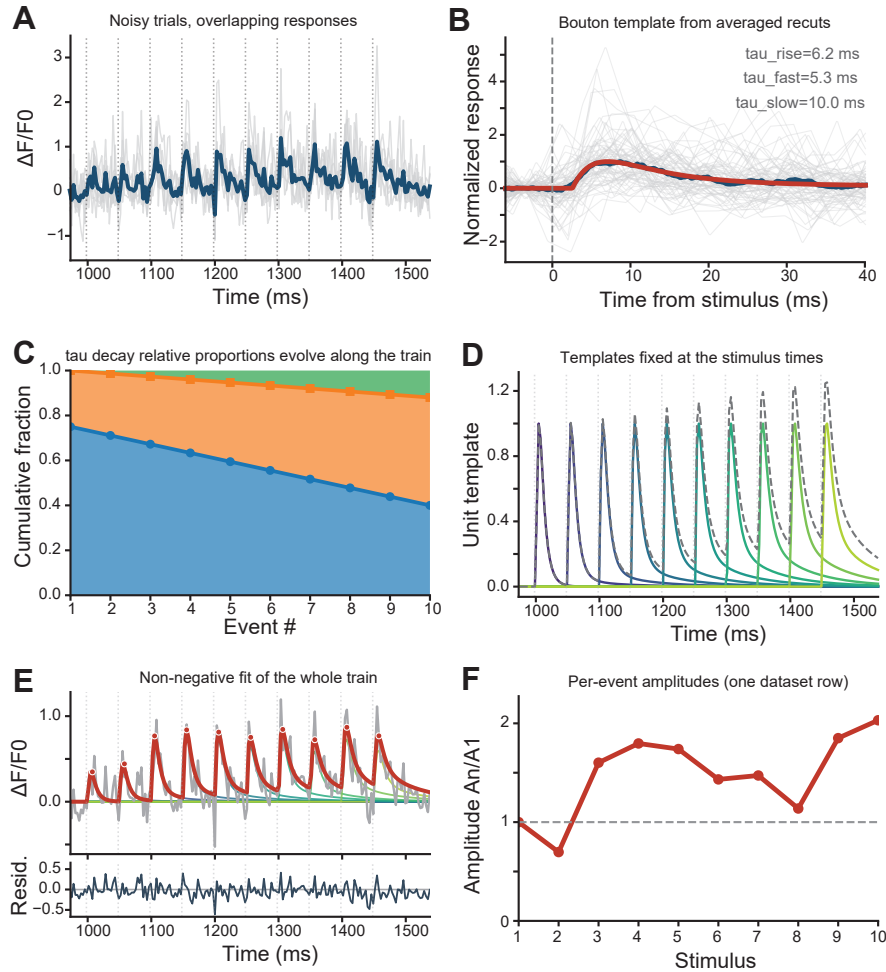


Fig. S2: Bouton-specific NNLS pipeline for extracting event-by-event glutamate-release amplitudes from overlapping iGluSnFR trains.

A. Representative $\Delta F/F_0$ traces from repeated trials at a single bouton during a 10-pulse train, illustrating the strong noise and temporal overlap between successive responses that make direct peak-by-peak measurements unreliable. The dark trace shows the trial average; vertical dotted lines indicate stimulus times.

B. Bouton-specific response template derived from stimulus-aligned recut transients pooled across trials on an example bouton. Grey traces show oversampled recuts, the blue trace shows the median recut waveform, and the red trace shows the fitted iGluSnFR template for one example bouton.

C. Schematic of the event-by-event evolution of template composition across the train. The relative contributions of the fast, slow, and super-slow components were allowed to vary smoothly between events, consistent with gradual slowing or accumulation during repetitive stimulation.

D. Unit templates placed at the known stimulus times to form the basis set used for deconvolution. Each colored trace corresponds to one event template; the dashed grey trace shows their overlapping sum.

E. Joint NNLS fit of the full fluorescence train. Colored traces show the contribution of each event, the grey trace is the bouton-average fluorescence signal, and the red trace is the summed NNLS reconstruction. Circles mark the extracted amplitudes of individual Glu-T, obtained from the underlying Savitzky-Golay-filtered trace at the NNLS peak time. The residual, raw trace minus NNLS fit, is shown below.

F. Per-event amplitude profile for the same bouton, plotted as amplitudes normalized to the first response, A_n/A_1 , yielding the short-term plasticity profile used for downstream analyses.

Fig S3

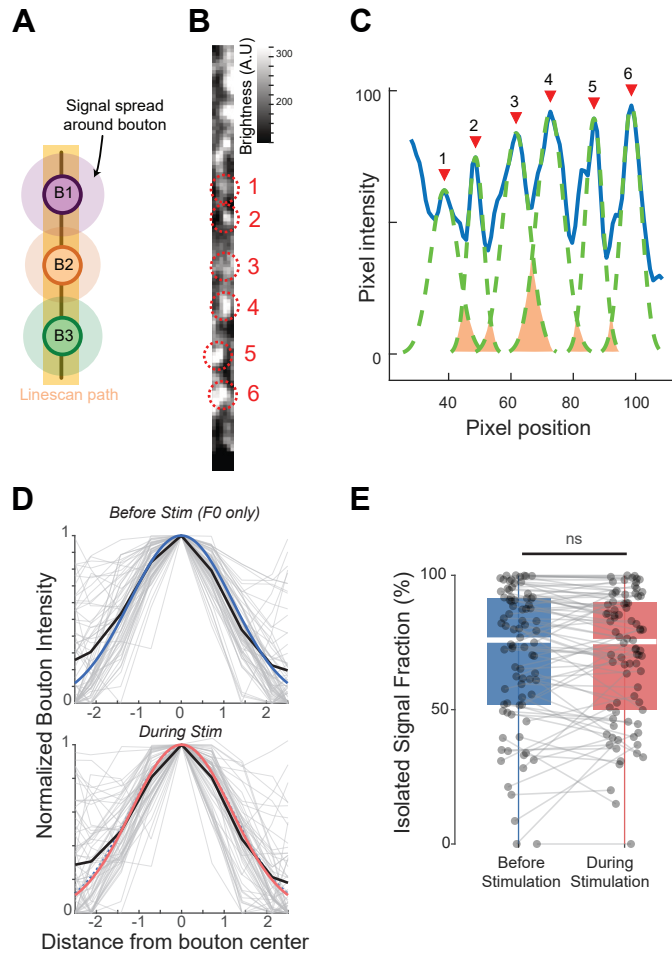


Fig. S3: Crosstalk between consecutive boutons along the linescan path is limited.

A. Schematic illustrating the experimental geometry: a single linescan path crosses three consecutive boutons, B1, B2, and B3. Colored discs around each bouton represent the lateral extent of the detected fluorescence from each central bouton, due to recording-related sources, e.g., the PSF, or biological sources, e.g., spillover. **B.** Example fiber, shown as a maximum projection during one trial, with hotspots at six putative boutons. **C.** Pixel intensity profile obtained by projecting the linescan columns (blue), with peaks corresponding to the six boutons identified in B (red triangles, numbered 1–6). For each bouton, a Gaussian was fitted to the intensity peak (green, dashed), and the overlap between adjacent Gaussians was calculated (orange) to estimate the contribution of signal from neighboring boutons. The isolated signal fraction for each bouton corresponds to the area under the Gaussian minus the area of overlap with its neighbors. Boutons located at the extremities of the recording, having only one visible neighbor, were excluded from this quantification. **D.** Lateral extent of individual bouton fluorescence signal relative to the bouton center (individual boutons in grey) before (top, average in blue) or during glutamate release (bottom, average in red); $n=82$ boutons. **E.** Boxplot of the per-bouton signal index, defined as the percentage of the area under the curve covered by neighboring boutons, before, blue, baseline, and during, red, stimulation trains, $n = 82$ paired boutons. Each grey dot is one bouton, before stimulation. Whisker bounds: minima/maxima, median, interquartile range. Wilcoxon test: $p = 0.2866$. Boutons are typically with an isolated signal fraction around 75%.

Fig S4

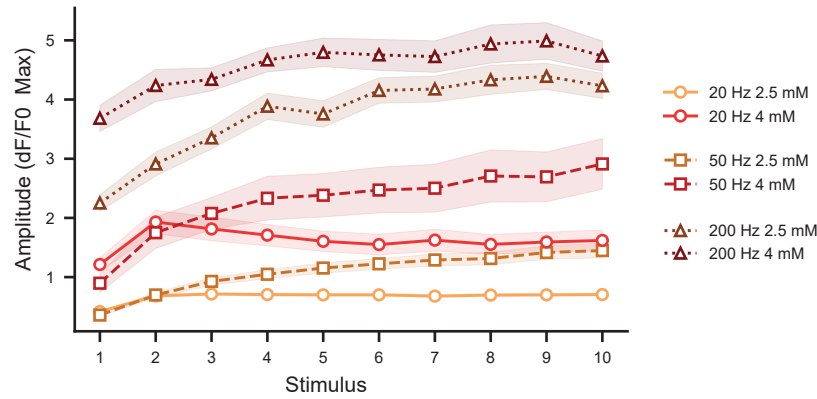


Fig. S4: SF-iGluSnFR.S72A operates within its non-saturating range under our experimental conditions.

Pulse-by-pulse uncorrected $\Delta F/F_0$ amplitudes across regular 20 Hz and 50 Hz trains and high-frequency 200 Hz saturation recordings at 2.5 mM and 4 mM $[Ca^{2+}]_e$. Lines and markers indicate mean responses; shaded areas indicate $\pm$ SEM. Group sizes: 20 Hz 2.5 mM, $n = 206$ boutons; 50 Hz 2.5 mM, $n = 44$ boutons; 200 Hz 2.5 mM, $n = 24$ boutons; 20 Hz 4 mM, $n = 21$ boutons; 50 Hz 4 mM, $n = 16$ boutons; 200 Hz 4 mM, $n = 27$ boutons. The 200 Hz saturation responses remained on average above all other responses for all pulses, at all frequencies and all conditions.

Fig S5

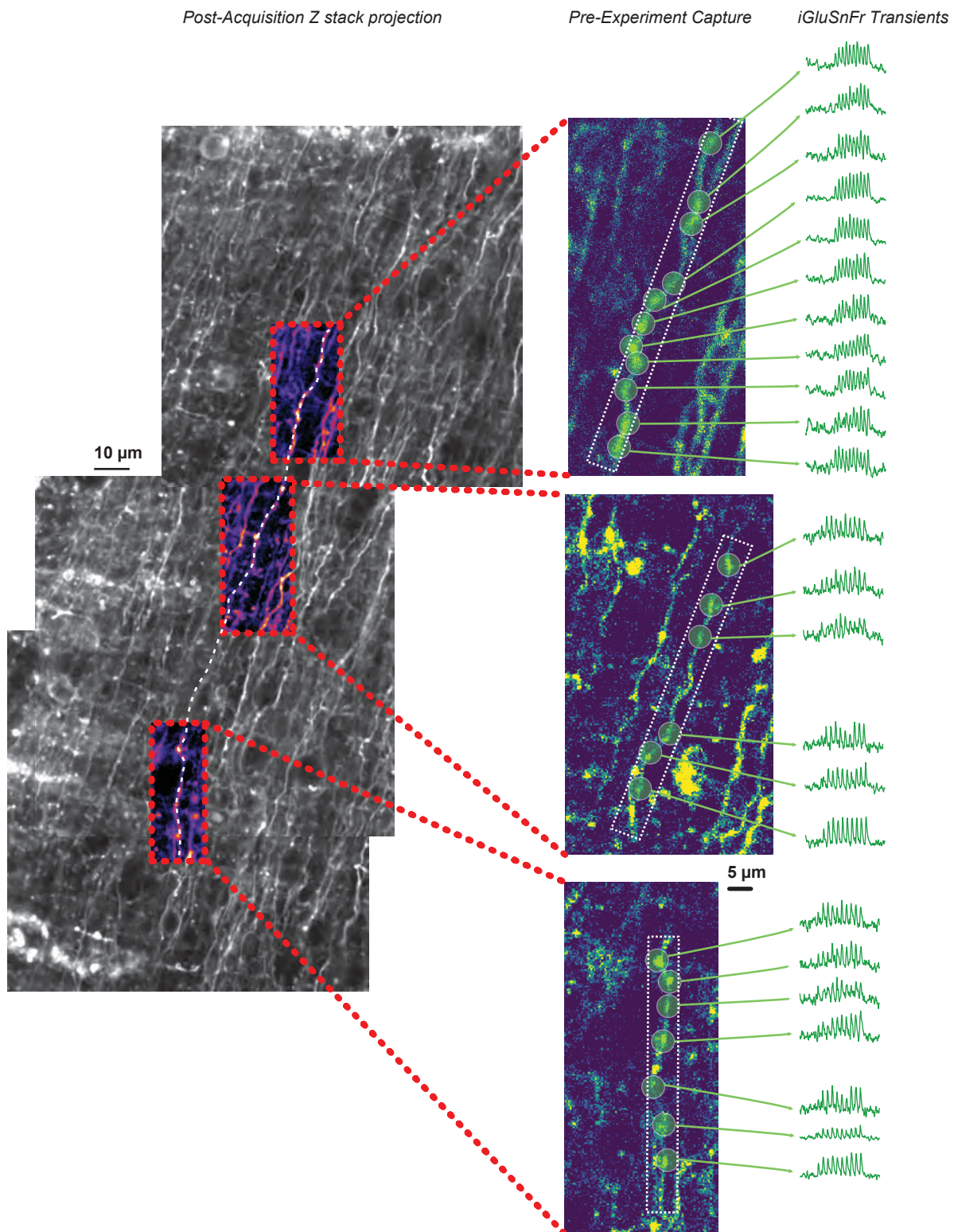


Fig. S5: Imaging extended axonal segments potentially spanning a microzone boundary.

Overview of three extended parallel fiber segments expressing SF-iGluSnFR.S72A. Left, greyscale images show median projections from final anatomical z-stacks acquired after the functional imaging experiment. The long white dotted line indicates the axonal segment followed across the three fields of view. Red dotted boxes mark the regions surrounding the functional imaging area. The pseudocolor images are single-plane fluorescence images acquired from the stack. Middle, single-plane fluorescence images acquired during the experiment, before substantial bleaching of the fiber could occur during the final z-stack acquisition. The white dotted line indicates the location of the imaging patches, typically made of six linescans. Green circles mark selected fluorescent boutons along each axonal segment. Corresponding $\Delta F/F$ traces are shown on the right for boutons connected by green lines, illustrating stimulus-evoked glutamate transients during 20 Hz extracellular parallel fiber stimulation at 2.5 mM extracellular $[Ca^{2+}]$.

Fig S6

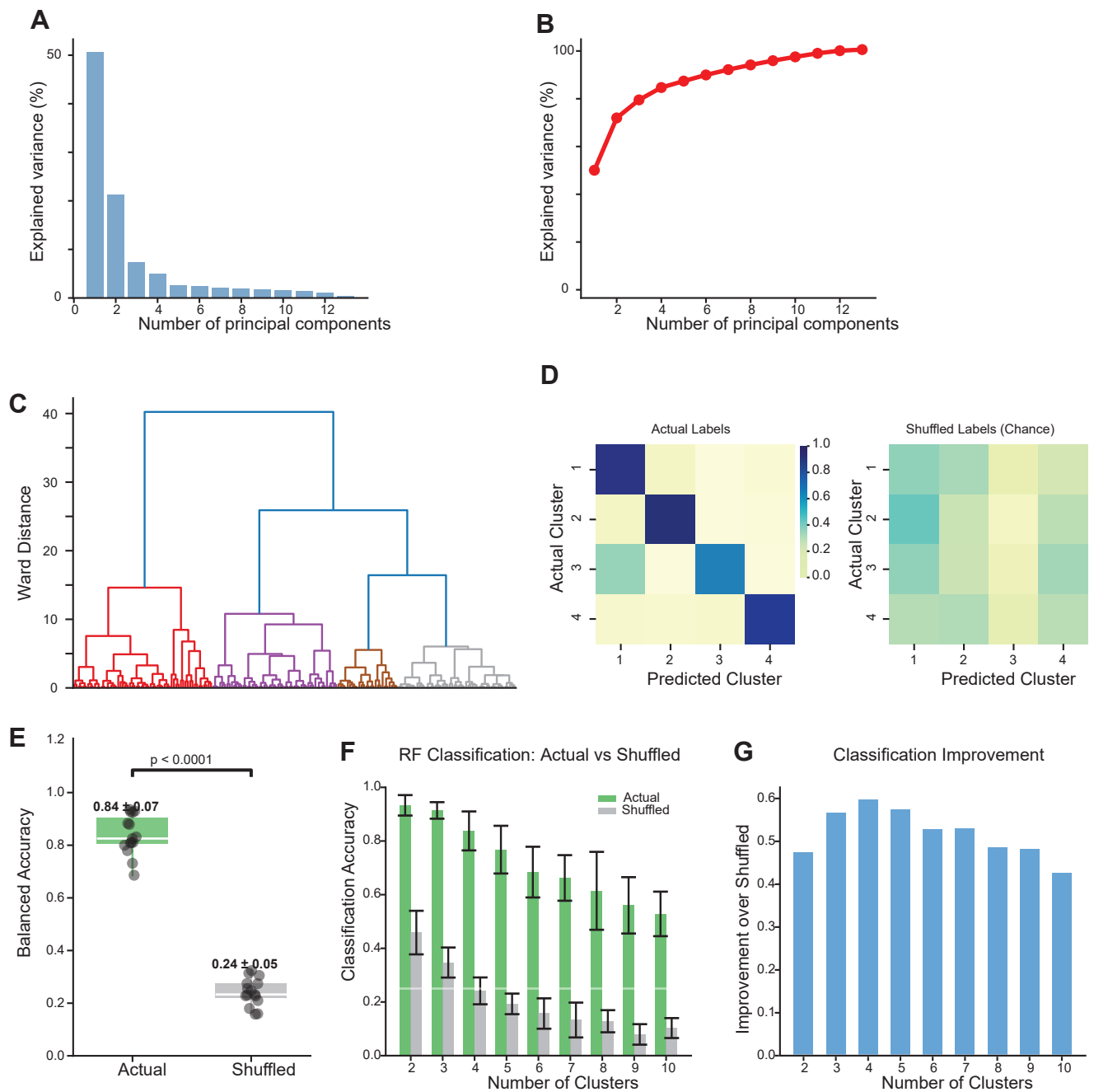


Fig. S6: Dimensionality, clustering, and supervised validation of the WT release-profile space.

A. Explained variance for each principal component obtained from PCA of the 13-feature WT 2.5 mM, 20 Hz reference dataset. **B.** Cumulative explained variance as a function of the number of principal components; PrC1 and PrC2 together account for 72.1% of the total variance. **C.** Dendrogram from hierarchical Ward clustering of WT boutons in the PrC1/PrC2 plane. Branch colors indicate the $k = 4$ cluster partition used throughout the manuscript: C1, $n = 51$; C2, $n = 60$; C3, $n = 26$; C4, $n = 54$. **D.** Mean confusion matrices from a class-balanced random-forest classifier trained to predict the four WT cluster labels from the 13 PCA input features, using stratified 5-fold cross-validation repeated three times. Left, actual cluster labels; right, randomly shuffled labels. **E.** Balanced accuracy for actual labels, green, was 0.84 ± 0.07 , compared with 0.24 ± 0.05 for shuffled labels, grey. All actual-label cross-validation scores exceeded the shuffled-label controls. **F.** Classification accuracy as a function of the number of clusters, from $k = 2$ to $k = 10$, for actual labels, green, and shuffled labels, grey. Bars and error bars indicate mean $\pm$ SD across cross-validation scores. **G.** Classification improvement, defined as actual minus shuffled balanced accuracy, as a function of cluster number. The improvement is maximal at $k = 4$ and decreases for larger cluster solutions.

Fig S7

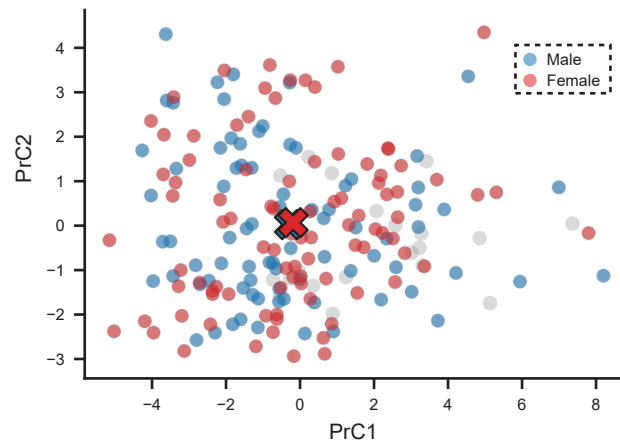


Fig. S7: GC boutons from male and female mice are equally distributed across the WT PCA release-profile space.

WT GC boutons from male, blue, and female, red, mice projected onto the PrC1/PrC2 space of Fig. 2: $n_{\text{male}} = 79$ boutons, $n_{\text{female}} = 92$ boutons; in grey, 20 unlabelled recordings. Red and blue crosses indicate the centroids of each population. Centroids overlap, and no significant separation between sexes is observed across PrC1 or PrC2: PrC1, $p = 0.43$; PrC2, $p = 0.85$.

Fig S16

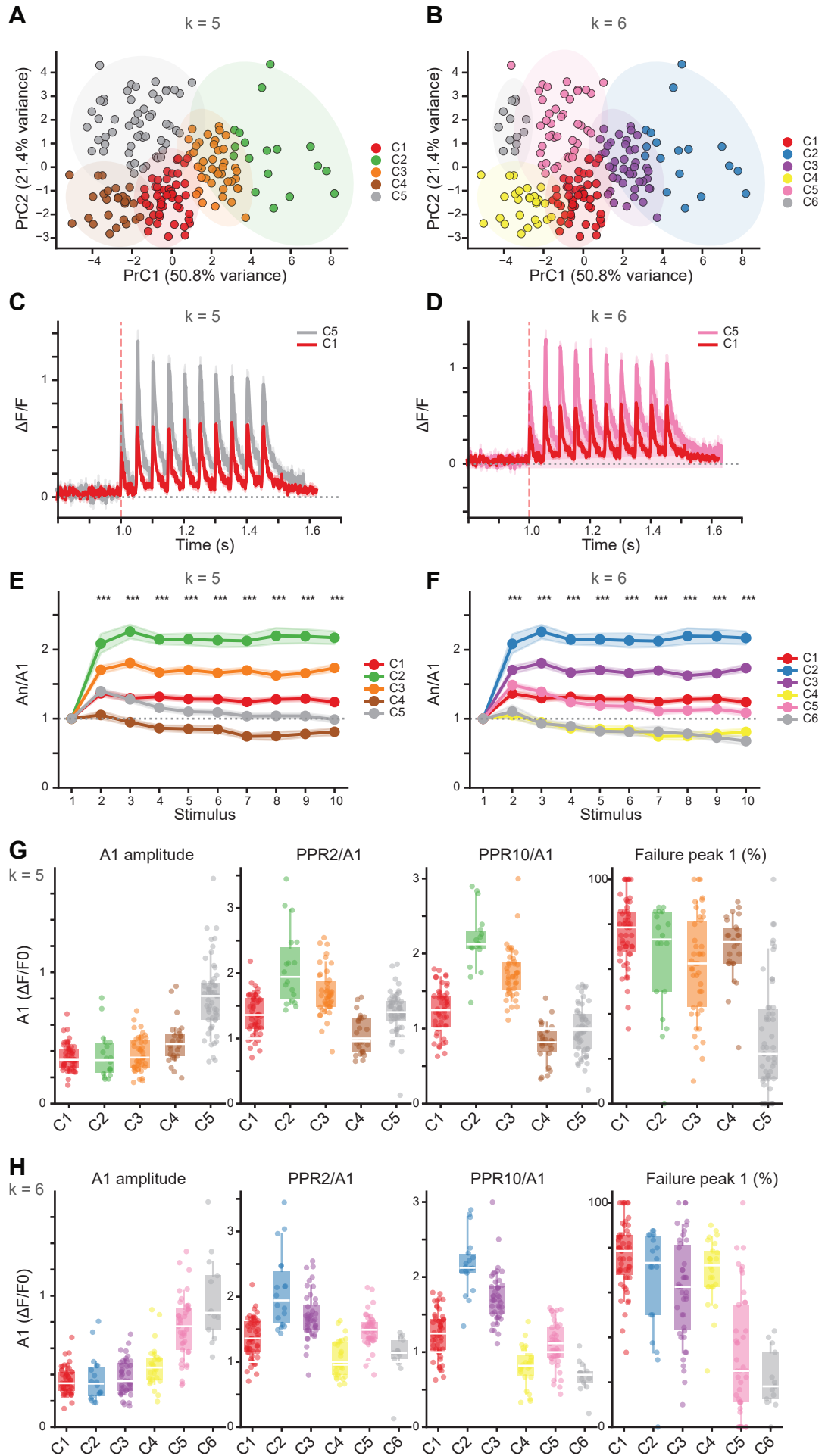


Fig. S8: Partitioning the bouton population into five or six clusters subdivides, but does not reshape, the four-cluster solution.

To test whether the four-cluster description of Fig. 2 depends on the chosen number of clusters, the same unsupervised pipeline (PCA on the standardized release-feature set followed by hierarchical clustering; $n = 191$ boutons) was re-run while forcing five ($k = 5$) or six ($k = 6$) clusters. Over-partitioning only subdivided existing clusters: no qualitatively new release profile appeared, and the dissociation between initial synaptic strength (A_1) and short-term dynamics ($PPR_{2/1}$) that motivates the analysis was present at every k . Cross-tabulation of single-bouton assignments confirmed strict nesting relative to the four-cluster solution ($C1 = 51$, $C2 = 60$, $C3 = 26$, $C4 = 54$): the $C2$ cluster split into $n = 18$ and $n = 42$ at $k = 5$, and the high-strength $C4$ cluster split into $n = 41$ and $n = 13$ at $k = 6$, while the $C1$ ($n = 51$) and depressing $C3$ ($n = 26$) clusters were preserved unchanged. In each panel pair the left panel shows $k = 5$ and the right panel $k = 6$.

A and B. Projection of all boutons onto the PCA release-profile space defined by $PrC1$ and $PrC2$, each bouton colored by cluster assignment, for $k = 5$ (A) and $k = 6$ (B), as in Fig. 2B. **C and D.** Mean $\Delta F/F_0$ traces for $C1$ and $C5$ in response to 10 pulses at 20 Hz, 2.5 mM $[Ca^{2+}]_e$, for $k = 5$ (C) and $k = 6$ (D), as in Fig. 2E; each trace is the mean cluster-specific fluorescence profile, shaded areas $\pm$ SEM, vertical ticks indicate stimulus times. **E and F.** STP profiles (A_n/A_1) across the train for each cluster, for $k = 5$ (E) and $k = 6$ (F), as in Fig. 2F; dots and lines indicate cluster means, shaded areas $\pm$ SEM. Non-parametric profile analysis: $k = 5$, overall Kruskal–Wallis $H = 153.8$, $p = 3.2 \times 10^{-32}$; permutation group $\times$ pulse test, $p = 2.0 \times 10^{-4}$; $k = 6$, overall Kruskal–Wallis $H = 164.9$, $p = 9.0 \times 10^{-34}$; permutation group $\times$ pulse test, $p = 2.0 \times 10^{-4}$. Pulses 2–10 differed across clusters at both k (pulse-by-pulse Kruskal–Wallis tests, FDR-corrected). **G and H.** Boxplots of A_1 amplitude, $PPR_{2/1}$, $PPR_{10/1}$ and first-pulse failure rate ($\%Fail_1$) across clusters, for $k = 5$ (G) and $k = 6$ (H), as in Fig. 2G. Kruskal–Wallis across clusters: $k = 5$ ($df = 4$) — A_1 $H = 91.7$, $p = 5.6 \times 10^{-19}$; $PPR_{2/1}$ $H = 69.1$, $p = 3.6 \times 10^{-14}$; $PPR_{10/1}$ $H = 117.5$, $p = 1.8 \times 10^{-24}$; $\%Fail_1$ $H = 74.8$, $p = 2.2 \times 10^{-15}$. $k = 6$ ($df = 5$) — A_1 $H = 92.8$, $p = 1.7 \times 10^{-18}$; $PPR_{2/1}$ $H = 78.7$, $p = 1.6 \times 10^{-15}$; $PPR_{10/1}$ $H = 125.4$, $p = 2.3 \times 10^{-25}$; $\%Fail_1$ $H = 76.0$, $p = 5.8 \times 10^{-15}$.

Critically, the strength/dynamics paradox persisted under over-partitioning. At $k = 5$, clusters $C1$ and $C5$ differed strongly in first-pulse amplitude yet were statistically indistinguishable in $PPR_{2/1}$ (Dunn/FDR: A_1 $p = 5.3 \times 10^{-16}$; $PPR_{2/1}$ $p = 0.70$). At $k = 6$ the same dissociation recurred for $C1$ and $C5$ (A_1 $p = 1.8 \times 10^{-12}$; $PPR_{2/1}$ $p = 0.14$). The amplitude–PPR dissociation is therefore independent of the chosen cluster number, indicating that it is a genuine feature of the data rather than an artifact of the four-cluster partition, consistent with the two-dimensional organization of release properties along $PrC1$ and $PrC2$.

Cluster sizes: $k = 5$, $C1 = 51$, $C2 = 18$, $C3 = 42$, $C4 = 26$, $C5 = 54$; $k = 6$, $C1 = 51$, $C2 = 18$, $C3 = 42$, $C4 = 26$, $C5 = 41$, $C6 = 13$.

Fig S9

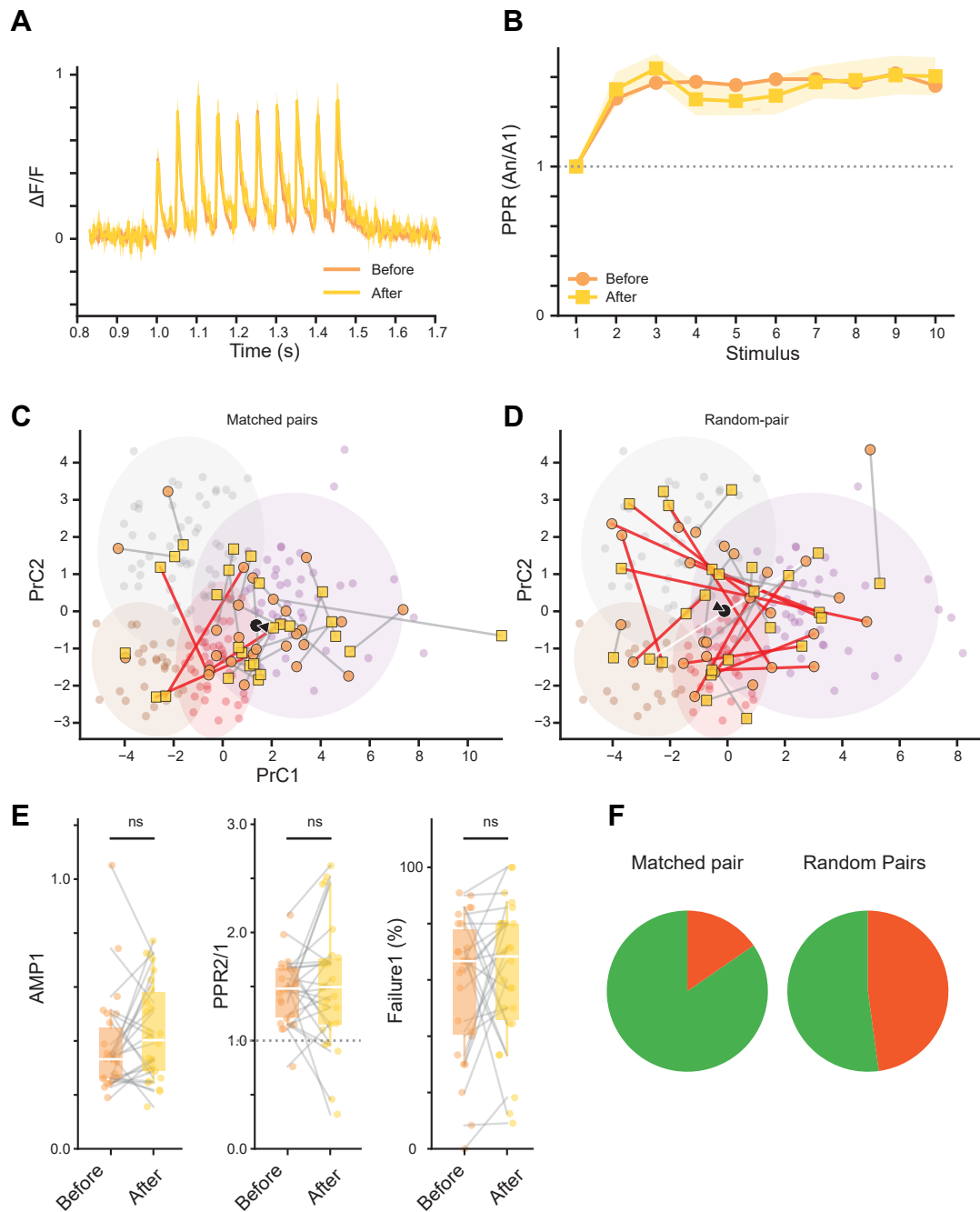


Fig. S9: Glu-T responses are stable in time and within cluster boundaries.

A. Single-bouton example: mean fluorescence trace at time 0, orange, “Before”, and after 8 min, yellow, “After”, in response to 20 Hz extracellular PF stimulation at 2.5 mM $[Ca^{2+}]_e$. Light traces, individual trials; dark traces, means; $n = 25$ paired boutons. **B.** A1 paired t test, $p = 0.198$; $PPR_{2/1}$ paired t test, $p = 0.562$; failure-rate Wilcoxon test, $p = 0.258$; paired $n = 25$. **C.** Displacement of matched bouton pairs in the WT PCA release-profile space, overlaid on the WT cluster ellipses, colored as in Fig. 2. Grey and red lines connect before/after pairs. Red indicates pairs that changed cluster identity. Black arrow, mean displacement vector, Mean Δ real. **D.** Same analysis as in C for a bootstrap chance-level reference obtained by drawing random bouton pairs from the full WT PCA cloud. Real matched-pair displacements are markedly smaller than random-pair displacements. **E.** Paired boxplots of A1, left, $PPR_{2/1}$, middle, and first-pulse failure rate, right, for matched boutons before and after 8 min. Grey lines connect paired boutons. A1 paired t test, $p = 0.198$; $PPR_{2/1}$ paired t test, $p = 0.562$; failure-rate Wilcoxon test, $p = 0.258$; paired $n = 25$. **F.** Proportion of matched bouton pairs whose two recordings remain compatible with at least one common WT cluster ellipse, stable. Real matched pairs: 84.6% stable, 22/26 pairs. Bootstrap chance level: 52.2% stable, 95% CI 47.4-56.7%.

Fig S9

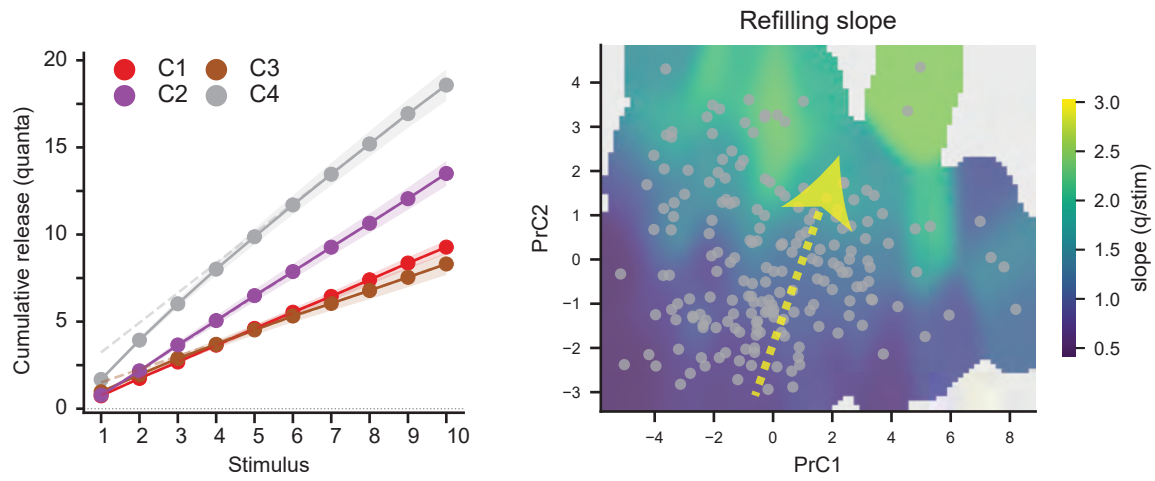


Fig. S10: Cumulative Release slope (replenishment rate) projection in PCA space.

A, Cumulative quantal release by group. Solid lines indicate means with SEM; dashed lines indicate steady-state fits to the final four pulses. **B**, Cumulative release slope, (replenishment rate, expressed in vesicles per stimulus), mapped onto the PCA space. The yellow arrow shows how boutons with increased replenishment rate value tend to progress along both PrC1 and PrC2.

Fig S10

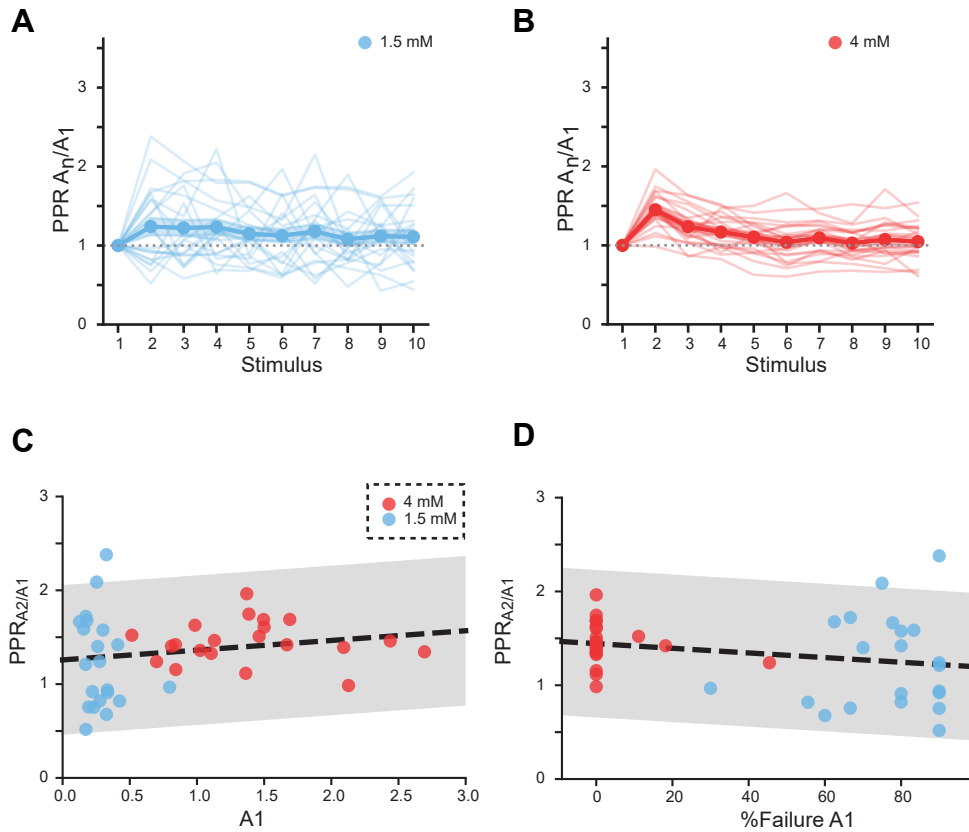


Fig. S11: $PPR_{A2/A1}$ is not predicted by initial A₁ amplitude or first-pulse failure rate across Ca^{2+} conditions.

A. Mean STP profile, A_n/A_1 , across all PF boutons recorded at 1.5 mM $[Ca^{2+}]_e$ and 20 Hz. Light blue traces, individual boutons, n = 21 boutons; blue trace and dots, average profile.

B. Mean STP profile, A_n/A_1 , across all PF boutons recorded at 4 mM $[Ca^{2+}]_e$ and 20 Hz. Light red traces, individual boutons, n = 21 boutons; red trace and dots, average profile.

C. Scatterplot of $PPR_{A2/A1}$ as a function of A₁ amplitude across the combined low- (1.5 mM, blue) and high- $[Ca^{2+}]_e$ (4 mM, red) range. Black line, linear fit across both conditions; grey shaded area, 95% confidence interval. Pearson $r = 0.18$, $p = 0.25$; n = 21 boutons. **D.** Same as C for $PPR_{A2/A1}$ versus %Fail1: $r = -0.24$, $p = 0.12$; n = 21 boutons.

Fig S11

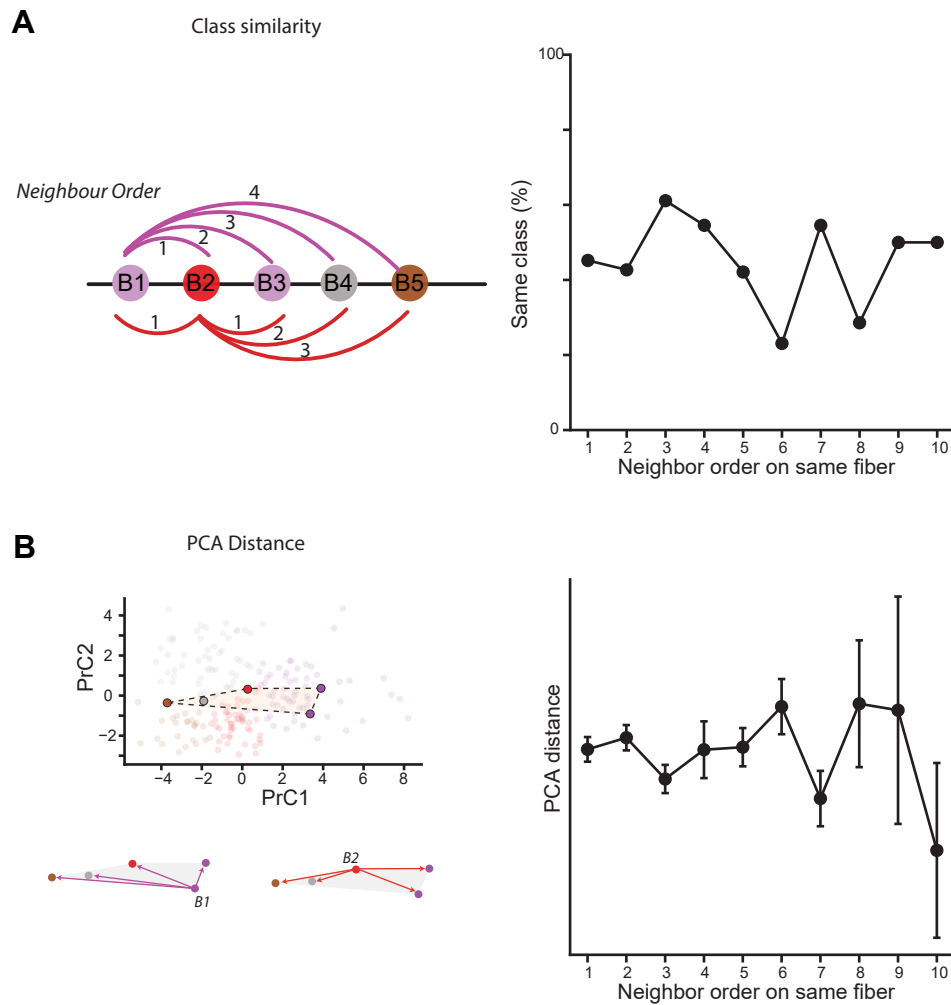


Fig. S12: Nearest-neighbor position does not predict release-class similarity along individual parallel fibers.

A. Left, schematic of the neighbor-order analysis along a single fiber. For each bouton, class similarity was evaluated with boutons at increasing positions on the same continuous fiber segment. Right, percentage of same-class bouton pairs as a function of neighbor order. Nearest-neighbor pairs were same-class in 45.2% of cases, N+1: 104 pairs, 33 fibers, and same-class probability did not differ significantly from this reference at any higher neighbor order after FDR correction. Fisher exact tests versus N+1; all FDR-adjusted $p \geq 0.597$. **B.** Left, schematic of the corresponding profile-space analysis in the WT PCA release-profile space. Dashed arrows indicate pairwise PCA distances between boutons from the same fiber. Right, mean PCA distance between bouton pairs as a function of neighbor order; error bars indicate SEM. Nearest-neighbor pairs had a mean PCA distance of 3.10 ± 0.19 , and no higher neighbor order differed significantly from this reference after FDR correction. Mann–Whitney U tests versus N+1; all FDR-adjusted $p \geq 0.562$. Contiguous same-segment bouton pairs were used for neighbor-order analysis, 329 pairs from 35 fibers. Cross-portion same-fiber pairs were excluded.

Fig S12

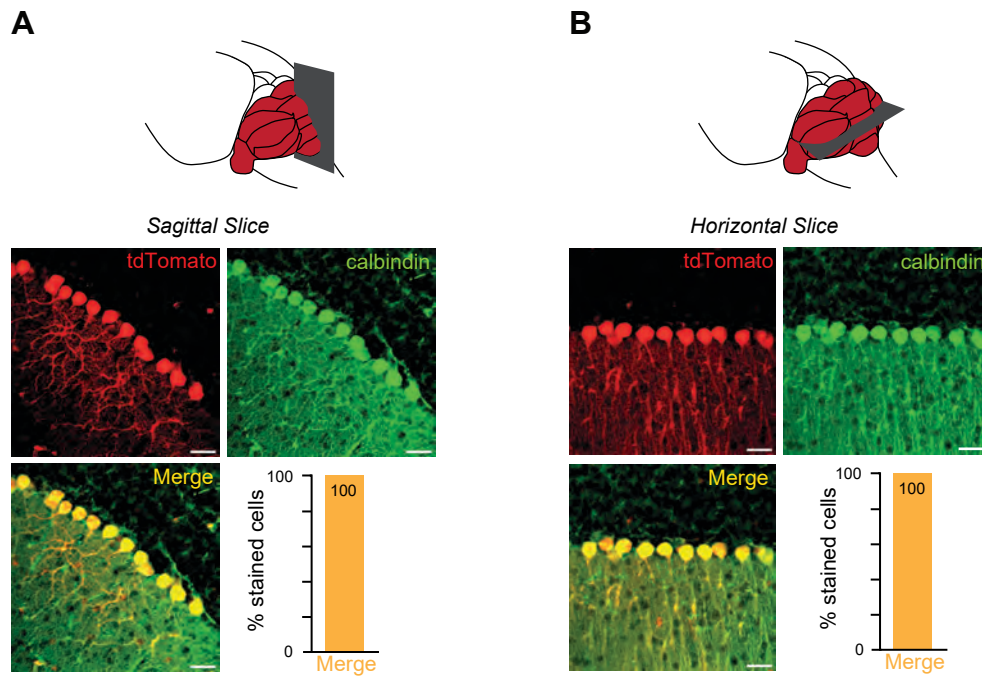


Fig. S13: All Purkinje cells in L7-tdTomato mice express tdTomato.

A. Top, sagittal sectioning plane. Bottom, confocal image of a sagittal cerebellar slice from an L7-tdTomato mouse showing tdTomato expression, red, calbindin D-28K immunostaining, green, and their overlay, Merge. All Purkinje cells co-express tdTomato and calbindin. Scale bars: 30 μ m. Bottom right, proportion of double-labeled cells: 100% Merge; sagittal, n = 112 cells; horizontal, n = 92 cells. **B.** Same as A in the horizontal plane.

Fig S13

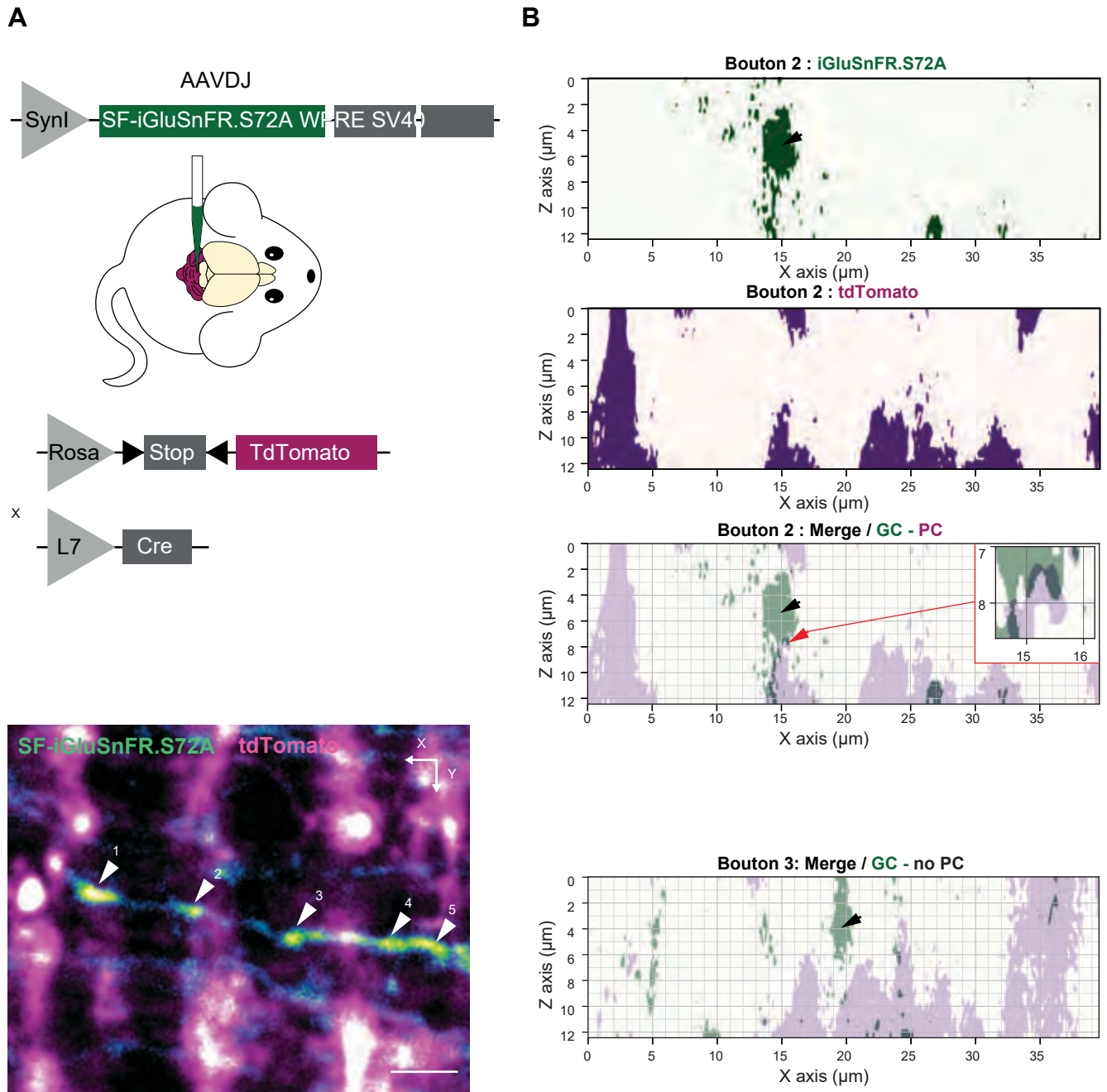


Fig. S14: Identification of PC versus non-PC postsynaptic targets.

A. Expression of SF-iGluSnFR.S72A in L7-tdTomato mice. Top, diagram illustrating viral injection of the AAV-DJ.hSynapsin.SF-iGluSnFR.S72A construct into the cerebellar vermis. Bottom, PFs expressing SF-iGluSnFR.S72A, green, and PCs expressing tdTomato, magenta. White arrowheads and numbers indicate boutons. Scale bar: 5 μm . **B.** Z-projection and Otsu binarization of SF-iGluSnFR.S72A, green, and tdTomato, purple, fluorescence signals. In the first example, the contact area between bouton 2, black arrowhead, and a PC dendrite is highlighted by overlap of the two channels, red arrowhead. Inset, magnification of the overlapping area. In the second example, bouton 3 does not overlap with PC dendrites. GC, granule cell; PC, Purkinje cell.

Fig S14

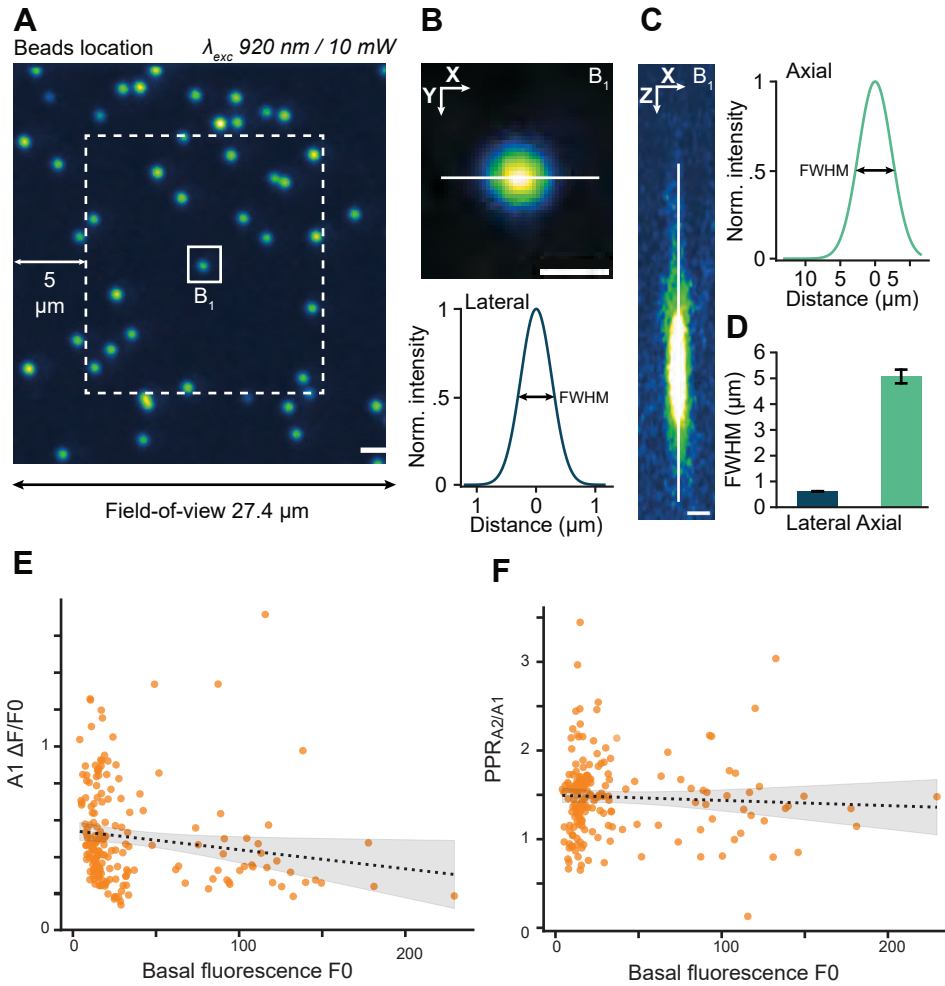


Fig. S15: Point-spread functions of the two-photon system and dependence of Glu-T amplitude on basal fluorescence

A. Average-intensity projection of a two-photon z-stack of fluorescent microbeads, 0.1 μ m diameter, λ_{exc} = 920 nm, 10 mW under the objective. The white dashed square indicates the central region from which microbeads were analyzed; microbeads within 5 μ m of the field-of-view edge were excluded. The microbead labeled B₁ is shown in B and C as an example. Scale bar: 2 μ m. **B.** Lateral fluorescence extension of B₁ in the XY plane. Top, magnification of B₁ bisected by a white line. Scale bar: 1 μ m. Bottom, normalized intensity profile along the bisecting line; full width at half maximum, FWHM, was used to define lateral resolution. **C.** Axial fluorescence extension of B₁ in the XZ plane, same conventions as in B. Scale bar: 1 μ m. **D.** FWHM lateral and axial resolutions across all analyzed microbeads. Mean $\pm$ SD: 0.62 $\pm$ 0.02 μ m for lateral resolution; 5.08 $\pm$ 0.86 μ m for axial resolution; n = 40 microbeads. **E.** Glu-T A1 amplitude as a function of basal fluorescence, F₀ showed low significant linear association with F₀: r = -0.16, p = 0.03; Dotted lines show linear fits and shaded areas show 95% confidence intervals. **F.** PPR_{A2/A1} as a function of F₀. Pearson correlations show no correlation; r = -0.053, p = 0.46; n = 191 boutons.

Fig S15

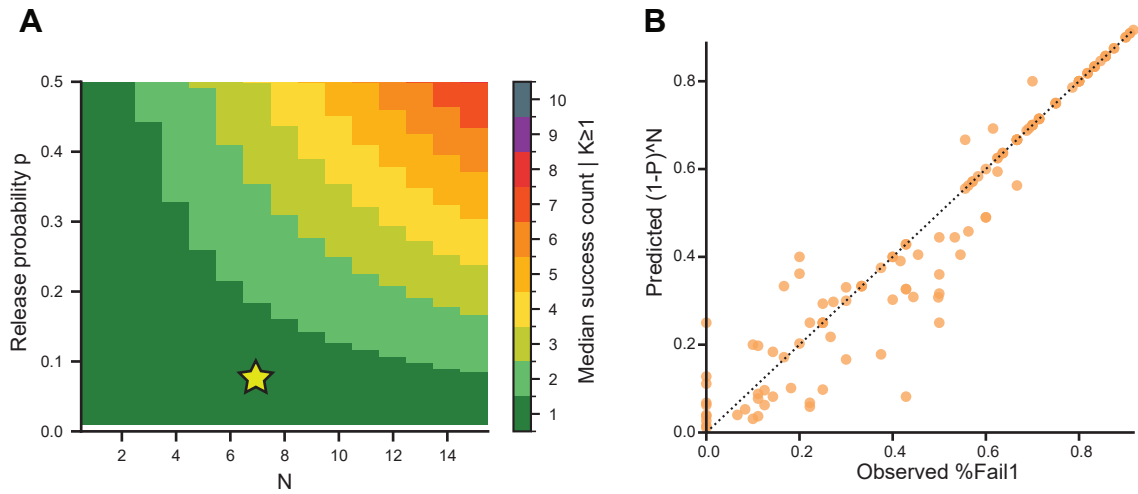


Fig. S16: Quantal screening and MLE failure-formula control.

A. Binomial N/P_r screening space showing the median number of released quanta among successful responses, $K \geq 1$, for $N = 1-15$ release sites and release probability $P_r = 0.01-0.50$. The color scale indicates the median success count, capped at 10 quanta, and identifies the parameter region in which successful responses remain effectively monoquantal. The star indicates the previously estimated average N and P values at this synapse (Valera et al, 2012) **B.** Comparison between observed first-pulse failures and the zero-release probability predicted from the MLE-derived N and P_r values, calculated as $(1 - P_r)^N$ for each bouton. The dotted line indicates identity. Observed and predicted failure probabilities closely matched across valid boutons: $n = 129$; observed %Fail1 mean, 49.1%; predicted $(1 - P_r)^N$ mean, 47.7%; mean $\Delta(\text{predicted} - \text{observed})$, -1.4 percentage points; Pearson $r = 0.961$.

Measure	Comparison	FDR-corrected p
A ₁ amplitude	C1 vs C2	0.647
A ₁ amplitude	C1 vs C3	0.00168
A ₁ amplitude	C1 vs C4	6.15e-15
A ₁ amplitude	C2 vs C3	0.0112
A ₁ amplitude	C2 vs C4	7.15e-15
A ₁ amplitude	C3 vs C4	3.92e-07
PPR _{2/1}	C1 vs C2	2.49e-07
PPR _{2/1}	C1 vs C3	1.29e-04
PPR _{2/1}	C1 vs C4	0.588
PPR _{2/1}	C2 vs C3	4.19e-10
PPR _{2/1}	C2 vs C4	2.42e-07
PPR _{2/1}	C3 vs C4	1.40e-05
PPR _{10/1}	C1 vs C2	8.24e-13
PPR _{10/1}	C1 vs C3	1.32e-06
PPR _{10/1}	C1 vs C4	2.50e-04
PPR _{10/1}	C2 vs C3	8.24e-13
PPR _{10/1}	C2 vs C4	9.48e-17
PPR _{10/1}	C3 vs C4	0.0235
Failure peak 1	C1 vs C2	0.00320
Failure peak 1	C1 vs C3	0.0709
Failure peak 1	C1 vs C4	2.76e-13
Failure peak 1	C2 vs C3	0.259
Failure peak 1	C2 vs C4	5.83e-09
Failure peak 1	C3 vs C4	1.91e-08

ns $p \geq 0.05$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Supp. Table 1. Pairwise comparisons between WT clusters for A₁ amplitude, PPRA₂/A₁, PPRA₁₀/A₁, and failure peak 1. P values are from two-sided Mann-Whitney tests with FDR correction following significant Kruskal-Wallis tests across the four clusters. Significance is indicated as ns, $p \geq 0.05$; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.